

Longitudinal magnetic resonance imaging studies of older adults: a shrinking brain.

Age-related loss of brain tissue has been inferred from cross-sectional neuroimaging studies, but direct measurements of gray and white matter changes from longitudinal studies are lacking. We quantified longitudinal magnetic resonance imaging (MRI) scans of 92 nondemented older adults (age 59-85 years at baseline) in the Baltimore Longitudinal Study of Aging to determine the rates and regional distribution of gray and white matter tissue loss in older adults. Using images from baseline, 2 year, and 4 year follow-up, we found significant age changes in gray ($p < 0.001$) and white ($p < 0.001$) volumes even in a subgroup of 24 very healthy elderly. Annual rates of tissue loss were 5.4 ± 0.3 , 2.4 ± 0.4 , and 3.1 ± 0.4 cm³ per year for total brain, gray, and white volumes, respectively, and ventricles increased by 1.4 ± 0.1 cm³ per year (3.7, 1.3, 2.4, and 1.2 cm³, respectively, in very healthy). Frontal and parietal, compared with temporal and occipital, lobar regions showed greater decline. Gray matter loss was most pronounced for orbital and inferior frontal, cingulate, insular, inferior parietal, and to a lesser extent mesial temporal regions, whereas white matter changes were widespread. In this first study of gray and white matter volume changes, we demonstrate significant longitudinal tissue loss for both gray and white matter even in very healthy older adults. These data provide essential information on the rate and regional pattern of age-associated changes against which pathology can be evaluated and suggest slower rates of brain atrophy in individuals who remain medically and cognitively healthy.